

Common rust

CAUSAL ORGANISM

Fungus: *Puccinia sorghi*

COMMON NAMES

Common rust of maize; rust of maize; common maize rust

HOST RANGE

Primary hosts:

Zea mays (maize) and *Zea mays* subsp. *mays* (sweet corn).

Wild hosts:

Oxalis corniculata (creeping sorrel).

IMPORTANCE

Common rust, as the name suggests, is the most frequently occurring rust in South Africa and all maize growing areas of the world. It reduces grain yield, primarily resulting from a reduction in kernel size. Estimates of losses vary, but are associated with the percentage of the leaf area infected at different growth stages. Estimates of reductions in grain weight range from about 3-8% for each 10% of the total leaf area infected. *P. sorghi* rarely causes significant yield reductions, however yield losses as high as 25% have been measured. Yields on some extremely susceptible hybrids may be reduced by as much as 75%.

DISEASE INFORMATION

Spores are wind blown with new infections occurring every 7 to 14 days. Moderate temperatures (16-25°C) and high relative humidity (>95%), associated with mist or dew for prolonged periods, favour rust development and spread. About 6 hours of moisture is required for infection, hence, leaves can often be infected even under relatively dry conditions, because tissue in the whorl is moist for the minimum period required for infection. Younger leaf tissue is more susceptible to infection than mature, adult leaves.

SIMILARITIES TO OTHER DISEASES

Common rust can sometimes be confused with Polysora rust caused by *Puccinia polysora*. Common rust pustules form on both upper and lower sides of an individual leaf, unlike Polysora rust, which predominately sporulates on the upper leaf surface. Pustules of Polysora rust are circular and light, cinnamon-brown in appearance and scattered over the whole leaf surface compared to common rust's darker, elongated pustules forming in a localised band pattern (refer to Polysora Rust fact sheet).



SYMPTOMS

Common rust is normally one of the first leaf diseases to be observed on maize in South Africa. Early lesions on leaves are small, pale, circular flecks occurring in clusters (Fig. 1). As lesions mature, it may be recognised by small, powdery pustules over both surfaces of the leaves. Brownish-red oblong pustules (Fig. 2) are the characteristic symptom on leaves in early stages of infection; later the fungus erupts (Fig. 3.) through the leaf surface (epidermis) exposing masses of powdery urediniospores. Spores produced within the pustules are easily dislodged and can be seen on a white shirt when walking through an infected field. Severe infections may form a rust band closer to the base of the leaf as a result of infection that took place when the leaf tissue was in the whorl. Band development causes death of the entire leaf blade (Fig. 4). As the season progresses these pustules darken as black overwintering teliospores are produced within the lesions (Fig. 5). Plants of the alternate host, the yellow flowering creeping sorrel (*Oxalis corniculata*) (Fig. 6), are subsequently infected producing light orange-coloured pustules called aecia (Fig. 7). This is simply another spore stage of the same fungus, a sexual stage responsible for the development of new races, which render current sources of resistance ineffective. Numerous physiological races of *P. sorghi* can be identified from reactions on maize lines with specific genes for resistance.

The red garden sorrel (*Oxalis latifolia*), with pinkish flowers, also gets infected by a rust towards the end of the maize season, showing an abundance of orange-coloured rust spores (Fig. 8). This rust is not to be confused with common rust and holds no threat to maize.

Common rust



Figure 1. Initial pin-point symptoms.



Figure 2. Brownish-red pustules.



Figure 3. Erumpent pustules.



Figure 4. Rust banding.



Figure 5. Black teliospores.



Figure 6. Creeping sorrel.



Figure 7. Aecia on *Oxalis corniculata*.



Figure 8. Rust on *Oxalis latiflora*.

CONTROL

Cultural control:

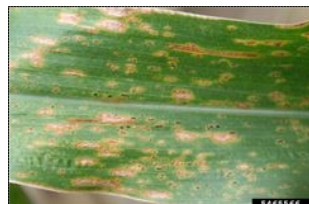
- Common rust is controlled effectively by planting resistant/tolerant hybrids (refer to the PANNAR Catalogue for information on resistance of PAN hybrids).
- Susceptible hybrids produce a magnitude of spores and can serve as a source of inoculum to neighbouring fields.
- Other than the alternate sorrel host, *P. sorghi* is very specific to maize and the spores cannot infect other crops e.g. beans, soybeans or sunflowers, therefore crop rotation will not really reduce risk of this disease.
- The disease is not seed- or soilborne and tillage practices will have minimal effect on disease control.

Chemical control:

- Timeous applications of fungicides - initial application when inoculum levels are low (a few pustules per leaf).
- In a suitable spray programme the first spray with one of the strobilurin/triazole combinations will act as a curative measure of the rust pathogen and also as a preventative to other leaf disease infections.
- The disease is not seedborne and seed treatments will not prevent common rust.

Common rust of Maize

Puccinia sorghi



Chlorotic areas developing on leaves (Daren Mueller, Iowa State University, Bugwood.org)



Red-brown pustules form on upper side of the leaf (Margaret McGrath, Cornell University, Bugwood.org)



Red-brown pustules of *P. sorghi* (Daren Mueller, Iowa State University, Bugwood.org)

Prevention

- **Use resistant varieties if available**
- Use clean maize seed which is from a reliable source
- Practise deep ploughing of crop residues (to at least 50 cm) if there is previous history of common rust in the field
- Practise crop rotation with non-cereal crops at least every 2-3 years (e.g. beans, fababean, chickpea).
- Remove and destroy weeds which are *Oxalis* spp. or *Zea* spp. as these are alternative hosts
- Plant maize in rows in the direction of prevailing winds for maximum air circulation

Monitoring

- Monitor the plant every week, especially when there are favourable environmental conditions (Mild (16-25 °C) and humid conditions (>95%) favour rust development)
- Start to observe the maize plant after 6 leaves emerge until the tasseling stage
- In early stages, small chlorotic areas develop on leaves. Later, small, elongated, scattered and dark red-brownish pustules form on both the upper and lower sides of the leaf
- Pustules may also appear on husks, tassels and stalks
- If the infection is severe then leaves and leaf sheaths turn yellow and die early

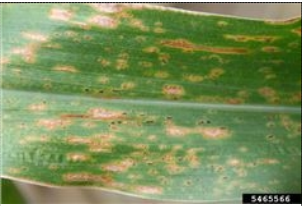
Direct Control

- Remove lower leaves which are infested to improve air circulation and reduce the inoculum (traces of fungi). Destroy the infected leaves by burning or burying deeply (at least 50 cm)

Note: Pesticides may be available to control this pest. Please check with the Ministry of Agriculture in your country to find out which pesticides are registered in your country and the local restrictions for their use.

Roya común del maíz

Puccinia sorghi

	Prevención	Monitoreo	Medida curativa
 Áreas cloróticas sobre hojas jóvenes en crecimiento (Daren Mueller, Iowa State University, Bugwood.org)  Pústulas de color herrumbroso oscuro que se forman sobre la superficie superior de la hoja (Margaret McGrath, Cornell University, Bugwood.org)  Pústulas de color herrumbroso oscuro de <i>P. sorghi</i> (Daren Mueller, Iowa State University, Bugwood.org)	• Usar variedades resistentes si están disponibles. • Practique incorporación profunda al suelo de los residuos de la cosecha. • Practique rotación de cultivos. • Remueva y destruya la maleza <i>Oxalis</i> sp., ya que es hospedero alternativo.	• Monitoree las plantas a cada cuatro semanas desde la etapa de cogollo hasta la etapa dentada y especialmente durante las dos semanas antes del espigamiento que es cuando las plantas son más susceptibles a la roya: • Durante las etapas tempranas de la enfermedad, pequeñas áreas cloróticas pueden ser observadas en las hojas. Después, pústulas café-rojizo pequeñas y alargadas se comienzan a distribuir en las superficies superior e inferior de las hojas. • Las pústulas también pueden aparecer sobre las tuzas, tallos y espigas. • Temperaturas entre 16-25°C y condiciones de alta humedad (>95%) favorecen el desarrollo de la roya.	

Note: Algunas plaguicidas pueden estar disponibles para el control de esta plaga. Consulte con el Ministerio de Agricultura de su país para saber qué plaguicidas están registrados en su país y las restricciones locales para su uso.

Gray Leaf Spot of Maize - Nepal

Cercospora zeae-maydis मकैको पातमा लाग्ने ध्वाँसे थ्रेग्ले रोग



Gray leaf spot infected leaf of maize (Photo: Gyanu Manandhar)



Gray leaf spot infected plants of maize (Photo: Gyanu Manandhar)

Prevention	Monitoring	Direct Control	Direct Control	Restrictions
- It is extremely important to remove plant debris from previous diseased crops. Rake up and burn the crop debris and to encourage your neighbours to do the same. The disease can easily spread from debris in your neighbours fields. The pathogen can survive on crop debris underground. - Use resistant/tolerant varieties like Ganesh-1, Manakamana-3 for high hills (1900 masl); and Manakamana-1, Deuti, Manakamana-6, ZM401 for mid hills (1300 masl). - Proper tillage practices as early planting as possible. - The pathogen can survive for long periods in the soil. Practice crop rotation for at least 3 years with non-host crops like millet, potato, vegetables, beans or pea. - Use recommended doses of chemical fertilizers (120:60:40 NPK kg/hectare). #REF!	- Monitor regularly for the appearance of gray leaf spot lesions on leaves just before or near initiation of tassel and ear shoot stage of plants grown at about 1400 masl. - Look for smaller chlorotic spots or lesions, first on lower leaves, which turn into long narrow rectangular lesions 2-3 weeks later. Damage to the stalks may cause plants to fall over. - Take action as soon as disease symptom is seen.	- To delay the disease, you can remove all the lower infected leaves at early stage of infection. Bury them at least 50cm deep away from field or burn. Practice this together with other adjoining farmers. - Avoid overhead irrigation.	- Use personal protection equipment while using fungicides. - Do not use chemicals with the same mode of action year after year as this can lead to resistance. - Always consult the most recent list of registered pesticides (PCPB). - Spray whole plant covering leaf, leaf sheath, stalk, ear shoot, as a protective measure, with carbendazim 50% WP @ 2 g/litre of water during first week of July or at initiation of tassel in disease-prone areas where susceptible varieties are grown. At least 35L of aqueous suspension of fungicide are required for 500 m2 area. - Soon after disease appearance, spray whole plant covering leaf, leaf sheath, stalk, ear shoot, with propiconazole 25 EC @ 1 ml/L of water. A repeated spray at the interval of 15 days may be required depending on the disease severity. At least 35 L of aqueous suspension of fungicide is required for 500 m2 area	- WHO Class: U (unlikely to present acute hazard). There is widespread resistance to this chemical and it should not be used in isolation. If used mix with Propiconazole (below) or similar 'azole' fungicide. A repeated spray at an interval of 15 days may be required depending on the disease severity. - Propiconazole falls in WHO list II (moderately hazardous). Has waiting period (PHI) of at least 15 days.

Nepal

CREATED/UPDATED: August 2015; Updated December 2023

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EDITED BY: Plantwise

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Gray Leaf Spot on Eggplant - Ghana

Stemphylium solani; *S. lycopersici* (syn. *S. floridanum*) Gray leaf spot disease of eggplant and pepper

	Prevention	Monitoring	Direct Control	Direct Control	Restrictions
 Spots on tomato leaf caused by <i>Stemphylium solani</i>. (Source: Clemson University - USDA Cooperative Extension Slide Series, Wikimedia Commons)  Reddish border of tomato leaf caused by <i>Stemphylium solani</i>. (Source: Clemson University - USDA Cooperative Extension Slide Series, Wikimedia Commons)  Gray leaf spot (<i>Stemphylium solani</i>). (Source: Florida Division of Plant Industry, Florida Department of Agriculture and Consumer Services, Bugwood.org)	• Avoid planting crops near other susceptible crops (e.g. tomato, pepper, eggplant). • Orient rows parallel to prevailing wind direction for rapid drying of leaves. • Avoid working in the field while the vegetation is wet. • Practice 3-year crop rotation with non-susceptible crops (i.e. non-solanaceous crops). • Remove crop residue from year to year. Eliminate weeds that can serve as intermediate hosts. • Avoid overhead irrigation as splashing of water can spread the disease. • Provide adequate ventilation for seedling beds. • Ensure good aeration in field using appropriate planting distances. • Water during the day more than in the evening.	• Look out for the following symptoms: • Small, circular to elongated brown spots with a reddish border on leaves, petioles, stems, fruit stalks and calyxes. • Red to brown spots with gray to white centre (frog eye), sometimes with yellow margins. • Elongated lesions on petioles and stems. • Leaf drop. • Presence of fungal fruiting bodies on the lesions, giving them a slightly light grey tint.	• When infection is low, remove affected leaves and destroy. • Treat with fungicides if disease intensity is high.	• Always read and follow manufacturer's instructions. • Wear protective clothing. • Rotate fungicides to prevent development of resistance. • Use fungicides only when disease intensity is high. • Spray chlorothalonil (75%) at 7-10 day intervals. Apply 90mls/15L water. • Spray mancozeb (80%) at 7-14 day intervals. Apply 85-100g/15L water. • Spray copper-based (Dispers) at 7-10 day intervals. Apply 100g/15L water.	• WHO Class U (unlikely to present an acute hazard), FRAC M5. PHI 7 days, REI 24hrs. • WHO Class U (unlikely to present an acute hazard), FRAC M3. Apply 3 times in a season. PHI is 5 days, REI 24 hrs. • WHO Class II (moderately hazardous), FRAC M1. PHI 5 days.

Ghana

CREATED/UPDATED: 08/02/2023

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EDITED BY: Plantwise



Leaf blight on Sorghum

Setosphaeria turcica Turcicum leaf blight, Northern leaf blight

	Prevention	Monitoring	Direct Control
 Large elliptical leaf blight lesions on sorghum (Courtesy EcoPort: David C. Nowell)  Typical shape and colour of a <i>S. turcica</i> lesion on a sorghum leaf (Courtesy EcoPort: David C. Nowell)  Early water-soaked lesion near the base of the leaf without a pigmented margin (Courtesy EcoPort: David C. Nowell)	• Use certified disease-free seeds. • Plant resistant or tolerant varieties. • To reduce initial inoculum, use tillage methods that promote the decomposition of crop debris, as the fungus survives in and on plant debris. • Practise crop rotation with non-grass crops and eliminate weedy grass hosts. • On sorghum, increased levels of nitrogen increase the development of the disease. Avoid high levels of nitrogen fertilization. • Crop rotation for 1-2 years with soybeans, beans and sunflower. Do not practise crop rotation with maize.	• Monitor regularly for symptoms during optimal conditions of high humidity and moderate temperatures (18-27 °C). • Leaves: • Initially slightly oval, small, water-soaked spots appear. The spots grow into elliptical (with pointed ends) brown lesions until nearly as wide as the leaf. • The margin is red-purple and during warm, humid weather, greenish-black fungal sporulation is produced in the centre of the lesions. • In the final stages, lesions are tan to grey, may merge and lead to complete burning of the foliage. • Lesions develop first on lower leaves and progress upwards. • Seedlings: small reddish-purple or yellowish-brown spots usually develop on the leaves. • Under warm, humid conditions the disease may cause serious damage by killing all leaves before plants have matured. • Symptoms on resistant cultivars are reduced to small chlorotic or necrotic spots. • <i>S. turcica</i> on sorghum forms larger and fewer lesions than <i>Cochliobolus heterostrophus</i> (southern leaf spot), mostly on the leaves but also on the stalks. <i>Cercospora</i>, <i>Colletorichum</i>, <i>Ascochyta</i> can also cause similar leaf disease symptoms. • Act when the first lesions appear on the leaves.	• There are no direct control measures for this disease.

Note: Pesticides may be available to control this pest. Please check with the Ministry of Agriculture in your country to find out which pesticides are registered in your country and the local restrictions for their use.



CABI PEST AND DISEASE PHOTOGUIDE TO

Sorghum disorders

KNOWLEDGE FOR LIFE

Introduction

This photo booklet has been produced by the CABI-led **Plantwise** programme (www.plantwise.org) to aid extension officers and other plant health advisors in diagnosing the most common pests, diseases and abiotic problems of sorghum around the world. The symptoms presented on a real plant sample can be compared with the photos in this guide to identify possible causes.

The booklet is organized into two broad sections, one showing the common insect pests that attack the crop and the other showing the various symptoms of poor health. In the symptoms section, the images are arranged by plant part, with similar-looking symptoms displayed together. Some biotic and abiotic factors cause more than one type of symptom, so there may be multiple images in different parts of the photo booklet for a specific problem. The photos for a particular problem are cross-referenced to make it easy to find all the relevant photos.

Contents

Sign or symptom	Box #
Insects	1–34
Leaf	35–57
Edible portion	58–70
Stems	71–83
Whole Plant	84–91
Root	92–98
Parasites	97–98

Beet armyworm

Spodoptera exigua

Worldwide except S America



Photos: M. Young, Rural Agricultural Development Authority, Jamaica; P Harris, Bugwood.org

- The larvae are pale green or yellow when small but gain pale stripes later.
- The body is practically without hairs or spines.
- Older larvae are generally darker but variable in colour, pink or yellow on the underside.
- The adults are moderately sized mottled grey brown moths, the wing span of 25-30 mm.

American cotton bollworm/Corn earworm

Helicoverpa zea

Americas and China



Photos: Clemson University – USDA Cooperative Extension Slide Series, Bugwood.org;
Seabrooke Leckie, Flickr

- Usually with orange heads and a black region just behind head.
- Bodies usually blackish but can be brown green/yellow covered with many thorny micro spines.
- Young larvae feed together but older ones eat each other; larger larvae are not so common.
- Adults variable colour (yellowish brown) 32-45 mm wingspan.

Oriental armyworm

Mythimna separate

Asia and SE Asia



Photos: Takahashi, Wikimedia commons; Shipher Wu, Flickr

- Young larvae are white, they turn green when older, head orange or brown.
- Mature larvae are variable but are greenish to light brown marbled orange brown with narrow white line along the back.
- Mature larvae bite off the grain stalks and let them fall.
- Adults about 20 mm long wingspan 40 mm forewings have two pale round spots.

Sorghum webworm

Nola sorghiella

N America only



Photos: Alton N. Sparks, Jr., University of Georgia, Bugwood.org; Andy Reago and Chrissy McClarren, Flickr

- External feeders of floral parts not leaves.
- Greenish/yellowish with 4 red/brown lines on back (max 13 mm) covered in spines they eat round holes in grains.
- When monitoring, shake heads of grain over white paper as larvae can be well hidden and not immediately obvious.
- Larvae spin cocoons.

BACK TO CONTENTS

Cotton bollworm

Helicoverpa armigera

Worldwide except N America



Photos: Paolo Mazzei, Bugwood.org; W. Billen, Pflanzenbeschaustelle, Weil am Rhein, Bugwood.org

- Up to 40mm long the head is yellow with several spots. Three dark stripes extend along the back and a yellow one along the side.
- Underside is pale.
- Larvae often eat each other; the older larvae are less common.
- When disturbed they drop from the plant and curl up.

African armyworm

Spodoptera exempta

Worldwide except S America



Photos: M Powell, ARC-PPRI; P Taylor, CABI

- Two types of larvae: those which like to live together and those which live alone.
- Grouped larvae have a velvety-black upper surface with pale lines on their sides, a green or yellow under surface, and no hairs on the body.
- There are three parallel lines on the back of the first body segment. The head is always shiny-black.
- Larvae living alone are fat and extremely sluggish and variable in colour.

Fall armyworm

Spodoptera frugiperda

America and Africa



Photos: Frank Peairs, Colorado State University, Bugwood.org; M. Cock, CABI

- Young larvae are greenish with a black head. Later the body becomes brownish, white lines develop down the sides.
- Raised spots are on the back; they are usually dark, and bear spines.
- The face of the mature larva is marked with a white upsided down “Y”.
- Adult moth: Triangular white spots at the tip and centre of the wings, rear wing is shiny silver-white with a narrow dark border round the edge.

Cutworm, Turnip moth larvae

Agrotis sp.

Worldwide



Photos: Frank Peairs, Colorado State University, Bugwood.org; Donald Hobern, Flickr

- The larvae are generally grey, sometimes tinged with purple.
- Live in the soil rarely climb plants.
- On each segment, 2 small black spots at the front and two at the back, each bearing a small bristle.
- Generally active at night will curl up if discovered during the day.

Sorghum chafer and other Chafer grubs

Pachnoda interrupta

Worldwide



Photos: P. Taylor, CABI; Udo Schmidt, Wikipedia

- Large fleshy body with brown head.
- Grubs always live underground and are characteristically C-shaped when disturbed.
- Three pairs of well-developed legs.
- Adults are typical beetle shaped, usually brown but may be patterned.

Wireworms, and false wireworm

Agriotes spp. *Gonocephalum* spp. and many others

Worldwide



Photo: Kansas State University

- Larvae are long and thin, small yellow brown, shiny hard-bodied and live in the soil.
- Can be move quickly with three pairs of well-developed legs near the head.
- Cannot curl into a tight C-shape as cutworms and chafer grubs do.

African sugarcane borer

Edlana saccharina

Africa



Photos: David Agassiz (top); ICIPE (lower)

- Oval yellow eggs. Larvae light-brown to dark-grey with small dark spots. Head reddish to dark brown.
- Often a great deal of frass hanging from exit holes.
- Moths vary considerably but have long forewings with two distinct dark spots at the top. When settled the sides of the wings are parallel to each other, 35 mm wingspan.

Sugarcane stalk borer

Diatraea saccharalis

N and S America



Photos: Aldo Norberto Bonaveri; William White, Bugwood.org

- Egg mass 25-50, whitish-yellow.
- Larvae 25 mm, red-brown head, yellowish-white body, brown spots.
- At low temp lacks spots and becomes yellower.
- Adults, straw-colour, V-shape black dots on forewing, 25 mm wingspan.

African maize stalk borer

Busseola fusca

Africa



Photos: D. Cugala ICIPE

- Eggs (white but turn darker with age) laid under leaf in a long column stretching up the stem.
- Young larvae are deep purple or black.
- Larvae featureless, up to 40 mm. The body is creamy-white with grey or pink hint, head dark brown.
- Adult wing-span 25-35 mm. The colour is variable depending on place and time of year.

BACK TO CONTENTS

Spotted Stalk borer

Chilo partellus

E Africa and S Asia



Photos: Rob Reeder, CABI; Georg Goergen/IITA Insect Museum, Cotonou, Benin

- Eggs 1.5 mm whitish and laid overlapping.
- Larvae up to 25 mm, reddish head, white-yellowish body, four dark stripes and spotted.
- Hooks on the prolegs (the four pairs of short stubby ones) are arranged in a circle (hand lens needed).
- The hooks of similar borers are arranged in a crescent or rows.

European corn borer

Ostrinia nubilalis

Worldwide except S America



Photos: Frank Pears, Colorado State University, Bugwood.org; Tony Morris, Flickr

- Eggs are nearly flat and 1 mm in diameter laid in a mass and overlap like tiles on a roof.
- The head is black in young larvae but in older ones they are dark brown.
- 4 large spots at the front of each segment and 2 smaller ones behind, small hairs grow from each spot.
- Underside is cream coloured and unmarked.

African pink stem borer

Sesamia calamistis

Africa



Photos: IRD Lab Evolution, Génomes et Spéciation; CIMMYT

- Eggs creamy white but darken with age, 1 mm wide.
- Larvae lack hairs and look creamy-white smooth and shiny. Often a distinctive pink colour under the skin.
- Hooks on the ends of short stubby legs are arranged in rows (hand lens needed).
- Unlikely to cause holing as it doesn't feed on the top leaves.

BACK TO CONTENTS

Purple stem borer

Sesamia inferens

S Asia and SE Asia



Photos: Agnes Adique Talavera courtesy Project Noah; IRRI

- Eats a wide range of crops, prefers rice not generally a major pest of Sorghum.
- The larva is without stripes but is obviously segmented.
- The legs are black and the last two segments have many black hairs
- Larval stage appears very similar to *Sesamia calamistis*.
- The adult moths have a thick brown hair tuft in the head and neck region.
- The wingspan up to 35 mm The forewings are light brown with some scattered dark brown spots throughout.

Earhead bug

Calocoris angustatus and other

India and Africa



Photo: Entomart, commons.wikipedia.org

- Eggs (blue) cigar shaped 1.5 mm laid under the scales covering the grain.
- Nymphs and adults attack developing grain (adults 5 mm long yellowish green).
- Nymphs and adults suck juice from grains when they are at the milky ripe stage.
- Grains fail to fill and heads become chaffy.

Southern green stink bug

Nezara viridula

Worldwide



Photo: R Ottens, University of Georgia, Bugwood.org

- Large and green (sometimes brown) 15 x 8 mm.
- Mostly a pest of legumes but can attack sorghum.
- Eggs are in groups of about 60. They are cream to yellow but become orange before hatching they are and tightly glued to each other.
- Young nymphs are black, older ones green. Nymphs and adults feed by sucking sap from developing grains.

Chinch bug

Blissus leucopterus

N and S America



Photo: Kansas State University

- Eggs are yellow but turn red found at the base of plants behind lower leaf sheaths and on soil.
- Nymphs, dark red and have a white band across their bodies. Adults 5 x 1.5 mm They are mostly black with reddish legs and milky white wings.
- Adults produce a familiar foul odour when crushed.

BACK TO CONTENTS

Corn planthopper

Peregrinus maidis

Worldwide



Photo: ARS Corn And Soybean Research Unit – ARS-USDA

- The nymphs have no wings and are green to light brown about 3 mm long.
- They run rapidly when disturbed. Both adults and nymphs like to feed inside the funnel of the plant, particularly in young plants. As the plants grow, they move out onto the underside of the leaves.

BACK TO CONTENTS

Stalk eyed fly

Diasemopsis spp.

Africa



Photo: Ton Rulkens Flickr

- The maggots feed inside the stems or tillers of seedlings results in “dead hearts”.
- Adults small, 6 mm and come out in the evenings.
- Several different types all with eyes on stalks.
- Damage can be mistaken for the Sorghum shoot fly.

BACK TO CONTENTS

Sorghum midge

Stenodiplosis sorghicola

Worldwide



Photo: State of Queensland

- Eggs are 0.4 mm long, tube shaped, transparent and with a light-red tinge in the middle one egg per spikelet at flowering.
- Older maggots become red and 2 mm. Red juice will flow from infested seeds when squeezed.
- Adults are, deep-red, 3 mm long, with see through wings. They emerge in the early mornings and can be seen hovering around developing grain heads.

BACK TO CONTENTS

Sorghum stem fly

Atherigona soccata

Africa, Asia, SE Asia

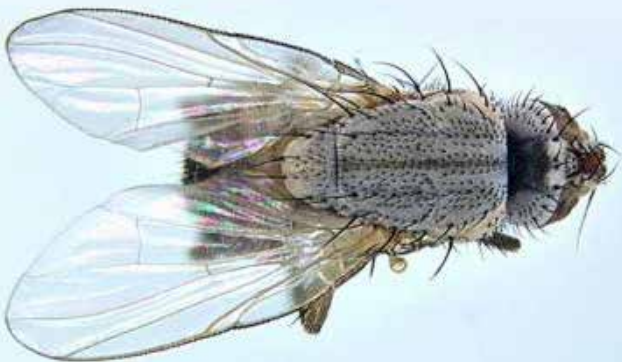


Photo: G. Goergen, Ecoport.org

- White eggs are long and thin, 0.8 x 0.2 mm 1-3 eggs laid per leaf.
- Larvae (maggots) crawl down into funnel and eat the youngest leaves of seedlings: Deadheart develops maggot up to 10 mm.
- Leaf easily removed and gives off bad smell.
- Fly (4 mm) looks like a small house fly.
- Older plants are more resistant.

Desert locust

Scistocerca gregaria

Africa and Asia



Photo: H Hubert, Flickr

- Solitary desert locusts colour is greyish or brownish, but males may become yellow when they mature. Males are 40-50 mm, females are 50-60 mm long.
- When they cluster together they turn black and develop orange spots.

Migratory locust

Locusta migratoria

Africa, Asia, SE Asia



Photo: P Rowley, Flickr

- Body length varying from 35 to 50 mm for males and from 45-55 mm for females.
- Mandibles (mouth parts) are blue. Wing cases are shining and long: 45-55 mm (males), 50-60 mm (females).
- Wings are colourless, except smoky tint and black veins.

Lesser grain weevil

Sitophilus oryzae

Worldwide



Photo: Udo Schmidt, Flickr

- 2 mm long with long snout, brown/black with four orange spots on wing cases.
- Usually a pest of stored products but can attack plants in the field.
- Female chews hole in stored grain and lays an egg the larvae develop inside the stored seed.

BACK TO CONTENTS

Greater grain weevil

Sitophilus zeamais

Worldwide



Photo: Gary Alpert, Harvard University, Bugwood.org

- 3 mm long but very similar to the lesser grain weevil.
- Four orange spots on wing cases but not as clear as in the lesser grain weevil.
- Pest of stored grain, larvae are found inside the stored seeds.

BACK TO CONTENTS

Grain moth

Sitotroga cerealella (shown on Maize)

Worldwide



Photo: Clemson University – USDA Cooperative Extension Slide Series, Bugwood.org

- A major pest of stored grain. Larvae not often seen as they develop within a grain.
- Eggs laid in cracks on grain in field but moths do survive in stores too.
- Larvae bore chamber into grain forming exit hole (above) for emerging adult.
- Adult small white moth wingspan 13–20 mm.

Mites

Tetranychus urticae (and others)

Worldwide



Photo: F.C. Schweissing, Bugwood.org

- Mites are extremely small and difficult to see (0.3 mm x 0.4 mm).
- There are many types but all create similar damage and characteristic webbing.
- Do not look for mites without a hand lens and do not look for mites on the dead areas of the plant.

BACK TO CONTENTS

Grey sugarcane mealy bug

Saccharicoccus sacchari

Worldwide



Photo: CABI

- Pink, of typical mealybug appearance, often covered with white wax up to 7 mm long.
- Egg shaped, fleshy and with a slightly segmented body no obvious head or bottom end.
- For most of their life cycle they live in the highly protected environment behind the leaf sheath.
- May attack underground as well above ground.

Spring green aphid

Schizaphis graminum

Worldwide



Photo: Frank Peairs, Colorado State University, Bugwood.org

- Feed on underside of leaf. Female 1.5-2.0 mm and typically pale green often with a stripe down the back.
- Fully grown they have the classic pear shaped body of aphids but young ones are smaller and with parallel sides.
- Winged forms develop when population are high or nearing the end of the season.

Sugarcane aphid

Melanaphis spp.

Worldwide



Photo: D Sekula Flickr

- Live on the lower surface of leaves.
- 1.8 mm long pale yellow or orange with black tail pipes and black feet
- Later in the season the aphids will leave the leaves move up onto the filling grain.
- Winged forms develop when population high.
- Sooty mould is extremely common with this pest.

Green Corn Aphid

Rhopalosiphum maidis

Worldwide



Photo: Kansas state university, Dept. of Entomology

- Typical aphid shape (2 mm) with black legs and black head and black bottom pale green grey body.
- Often in groups in the curl of the leaves and upper part of the stalk and may completely cover a large area. They may be found in large numbers down in the funnel of the plant.

BACK TO CONTENTS

Zonate leaf spot

Gloeocercospora sorghi

Worldwide



Photo: L Claflin, Kansas State University

- Red spots on lower leaves are the first symptoms.
- These spots develop into circles (mid leaf) or semi-circular if on the edge of leaves but they then elongate and join together.
- It can become severe in times of high temperature and humidity.

BACK TO CONTENTS

Anthracnose foliar blight

Colletotrichum sublineolum

Worldwide



Photo: Clemson University – USDA Cooperative Extension Slide Series, Bugwood.org

- Circular dark spots with red edges and a lighter middle.
- Older spots get a brown centre, (2–20 mm).
- Black fruiting bodies within the spots are an important thing to look out for.
- Whole leaves can be killed and plant may die before harvest.

BACK TO CONTENTS

Sooty stripe (NOT Sooty mould)

Ramulispora sorghi

Worldwide



Photo: David Nowell, Ecoport.org

- Small yellow spots are the first symptom.
- These become oval and develop a brown centre with a wide yellow edge.
- Spots can get really large and join up.
- Older spots have a lot of black spores appear as a “soot” covering the surface of the brown area.

Northern leaf blight

Setosphaeria turcica

Worldwide



Photo: David Nowell, Ecoport.org

- On seedlings, small red spots join, leaf wilts, and plants die.
- Starts on lower leaves or (less so) on stems, long, oval brown spots, matures to yellow-brown with purple borders; can be as wide as leaf.
- Will kill whole plant if conditions allow.

BACK TO CONTENTS

Rust

Puccinia purpurea

Worldwide



Photo: S Nelson, Flickr

- Purple-red spots on both leaf surfaces, parallel to veins, they mature to 'bumps', which break into reddish powdery masses (pustules).
- On the head stalk long brownish streaks develop.
- Later in the season black spots develop on leaves; this is the same problem but in a different phase.

Southern leaf spot

Cochilobolus heterostrophus

Worldwide



Photo: CIMMYT, Flickr

- Tan spots 2-6 mm wide and 3-22 mm long.
- Spots have yellow green or chlorotic border.
- When the seed head emerges a black felty mould may cover the affected seeds and ear rot may occur.

Grey leaf spot

Cercospora sorghi

Worldwide



Photo: P. Taylor, CABI

- Rectangular narrow tan/brown spots appear on mature leaves (12-50 mm) x (3-6 mm) turn grey with time.
- Individual lesions may coalesce, and under severe disease conditions, blight entire leaves.

BACK TO CONTENTS

Rough leaf spot

Ascochyta sorghi

Asia, Africa, N America



Photo: Clemson University Extension slide series, Bugwood.org

- Spots first appear 2-3 mm x 2 mm as reddish areas.
- Fruiting bodies are usually produced rapidly sometimes before the discolouration (on both sides of the leaf).
- Spots enlarge until they become 8-14 x 4-8 mm. Usually the centres become tan with a purple or red border but this depends on the variety.

Spring green aphid

Schizaphis graminum

Worldwide



Photo: Alton N. Sparks, Jr., University of Georgia, Bugwood.org

- Extensive red brown spotting of leaves develops and remains even once the aphids have been controlled.
- Whole plants at seedling stage can be destroyed and feeding on older plants causes lodging.

Downy mildew

Peronosclerospora spp. (on Maize)

Worldwide



Photo: Thomas Isakeit, Texas A&M University

- Leaves have yellow to white stripes from the base.
- Leaves are often narrower than usual, erect, and shredded.
A white, downy growth may develop on both surfaces of leaves.
- Plants may be stunted, and have an irregular seed set.

Mite damage

Mite damage (on maize)

Worldwide



Photo: S. Zukoff, Kansas State University, Department of Entomology

- Silvering of the leaf associated with webbing.
- Death of the leaf soon follows but mites are not be seen on the dead leaves.
- Mites are commonly found on the back of the leaf near the midrib and near the leaf sheath.

BACK TO CONTENTS

Downy mildew

Peronosclerospora sorghi

Worldwide



Photo: Thomas Isakeit, Texas A&M University

- White downy growth on underside of leaves. Later, narrow yellow striping which turn brown.
- White areas dry, brown, die appear shredded.
- Plants become stunted, fail to produce grain.

BACK TO CONTENTS

Sugarcane aphid

Melanaphis spp.

Worldwide



Photo: Kay Ledbetter, Texas A&M AgriLife Communications

- Discolouration of the leaves is extremely common.
- Leaf spots develop spots usually on underside of leaf.
- Whole leaf drying can occur in severe situations

BACK TO CONTENTS

Corn planthopper

Peregrinus maidis



Photo: ARS Corn And Soybean Research Unit

- Small (5 mm) and look little bit like aphids.
- Waxy nymphs and adults often seen in plant funnel, on leaf sheaths and underneath leaves.
- Large numbers (looking like aphid colonies) will stunt the plant, leaves become twisted, brown and the ear will fail to emerge.
- Transmits Maize moasic virus.

Shortage of Phosphate

Phosphate deficiency

Worldwide



Photo: Dr Prakash Kumar

- Severe stunting of the plant associated with dark green colour and purpling of the lower leaves.
- Other stress factors can produce similar symptoms but this is very characteristic of phosphate (P) shortage.

BACK TO CONTENTS

Shortage of Nitrogen

Nitrogen deficiency

Worldwide



Photo: Dr Prakash Kumar

- General yellowing of the plant usually associated with stunting.
- Symptoms first seen in the lower leaves, tip of the plant remains green.
- Yellow V-shape spreads down the centre of the leaf from the tip of the leaf to the stem.

Shortage of Potassium

Potassium deficiency



Photo: Dr Prakash Kumar

- The edges of the lower leaves are the first to show symptoms.
- Symptoms first seen in the lower leaves, tip of the plant remains green.
- Unlike nitrogen (N) shortage where the area closest to the centre of the leaf turns yellow in the case of potassium (K) shortage the edges of the leaf turn yellow and then brown.

Shortage of Calcium

Calcium deficiency

Worldwide



Photo: Dr Prakash Kumar

- The plants are stunted and the leaves fail to expand properly and they are often partially trapped by the lower leaf.
- The plant may develop a ladder like appearance with each leaf holding the one above it.
- In this photo ignore the leaf borer damage and slight necrosis.

BACK TO CONTENTS

Shortage of Sulphur

Sulphur deficiency

Worldwide



Photo: Dr Prakash Kumar and Dr Manoj Kumar

- General yellowing of the plant similar to nitrogen shortage but in the case of sulphur shortage the tip of the plant also turns pale yellow whereas when nitrogen (N) is short it will remain green.

BACK TO CONTENTS

Maize streak virus

MSV Symptoms shown on Maize

Africa



Photo: P. Taylor, CABI

- Long stripes on the leaves usually seen in the younger leaves first.
- Appears similar to other causes but the stripes caused by downy mildew are much wider and those due to iron deficiency are continuous and longer.

BACK TO CONTENTS

Maize dwarf mosaic virus

MDMV

Worldwide



Photo: International Maize and Wheat Improvement Center

- Symptoms depend on many factors, environment, cultivar, strain of virus etc.
- Symptoms can include “mosaic” “red leaf” “tan stripe” and “red stripe”.
- Mottle may develop on young leaves, chlorotic dots enlarge to form stripes.
- In cool climates, red streaks turn brown.
- Stunting is a common symptom.

BACK TO CONTENTS

Downy mildew

Peronosclerospora sorghi

Worldwide



Photo: International Maize and Wheat Improvement Center

- Early symptoms, pale leaves, white downy growth on underside.
- On older leaves wide yellow or white striping, and often develop brown spots.
- White areas dry, brown, die and appear shredded.
- Plants are stunted and fail to produce grain.

BACK TO CONTENTS

Shortage of Iron

Iron deficiency

Worldwide



Photo: Dr Prakash Kumar and Dr Manoj Kumar

- Long and continuous stripes on the leaves.
- Usually seen in the younger leaves first.
- Leaves can become almost totally yellow in extreme cases.
- Stripes caused by viruses are usually not continuous and downy mildew stripes are much wider.

BACK TO CONTENTS

Ergot, sugary disease

Claviceps africana

Africa and S America



Photo: G Odvody, Texas A&M University

- At harvest time the fungus will no longer be active.
- Once the sugary honey dew stops the heads appear black and contain no or few seeds.

BACK TO CONTENTS

Ergot, sugary disease

Claviceps Africana

Africa and S America



Photo: G Odvody, Texas A&M University

- The sugary honey dew that oozes from the developing ears may develop a whitish bloom as shown above.
- This is the characteristic of this disease.

BACK TO CONTENTS

Crazy top

Sclerophthora macrospora

Worldwide



Photo: Paul Bachi, University of Kentucky Research and Education Center, Bugwood.org

- Characteristic 'crazy top'.
- Early infection: leaf mosaic Severe: lower leaves, strap-like, leathery, roughened underside, with translucent chlorotic streaks and twisted.
- Stunted roots, plant pulls up easily.
- Related to Downy mildew.

BACK TO CONTENTS

Grain mould

Various fungi

Worldwide



Photo: Tom Allen, Mississippi State University Extension

- Variable symptoms due to the nature of the problem.
- Severe infection results in grains with pink, white, or black fungal mycelium and sporulating structures.
- May lead to toxin production making the sorghum dangerous to eat and storage problems.
- Compact ear heads are highly susceptible in wet weather.

BACK TO CONTENTS

Sorghum midge

Stenodiplosis sorghicola

(worldwide)



Photo: E Boa, CABI

- Grains fail to develop resulting in black shrivelled heads.
- The maggots and the adult midge will not be present at this stage as they only attack the developing grain and not the mature seeds.
- Empty pupae cases may be visible on top of spikelets and adults may be seen.

Spring green aphid

Schizaphis graminum

Worldwide



Photo: D Mock, Texas A&M University

- Once the plant begins to produce seed the aphids may move up onto the developing stalks of the grain.
- Seed head becomes splayed.
- Feeding on the developing grains prevents them from filling and they will die.
- Many aphids can produce similar symptoms and the individuals need to be examined.

BACK TO CONTENTS

Covered kernel smut

Sporisorium sorghi

Worldwide



Photo: L Claflin, Kansas State University

- Plants are infected at the seedling stage but show no symptoms until near maturity.
- Dark brown powdery masses are covered by a tough greyish white or brown membrane, this usually breaks at harvest time.
- Usually all the seeds in the head are affected.

BACK TO CONTENTS

Head smut

Sphacelotheca reiliana

Worldwide



Photo: R Shivas, PaDIL.gov.au

- No symptoms except stunting initially; smut replaces seed before emergence from the leaves.
- Developing head is enclosed in a thick whitish membrane but this ruptures before emergence.
- Dark brownish dusty masses from tangled parts of plant strands will blow around.
- Small leaves may develop within the smutted head.

Covered kernal smut

Sporisorium sorghi

Worldwide



Photo: University of Georgia Plant Pathology, University of Georgia, Bugwood.org

- Infected grains turn white, uninfected ones remain brown.
- In this case seed head is only partially infected but usually all the seeds in the head are affected.

BACK TO CONTENTS

Loose kernel smut

Sphacelotheca cruenta

Worldwide



Photo: P Sajjabbi, www.runetwork.org

- Plants become infected as seedlings, plants are often stunted.
- Infected heads emerge before uninfected ones.
- Individual seeds are replaced by pointed galls (25 mm or longer), with a thin grey membrane.
- This membrane soon breaks after ear emergence and black dusty material is released and will blow away, long black pointed structures remain.

Ergot, sugary disease

Claviceps africana

Worldwide



Photo: G Odvody, Texas A&M University

- Clear sticky honeydew, oozes from grain and drips on leaves grain heads often compacted.
- May become blackened with sooty mould or may develop a whitish bloom on the honey dew surface.
- Black fungal survival structures produced later in place of the seed.

BACK TO CONTENTS

Sorghum midge

Stenodiplosis sorghicola

Worldwide



Photo: G Cronholm, Texas A&M University

- Maggots feeding on milky stage grains will make empty chaffy spikelets as shrivelled grains fail to develop.
- Empty pupae cases (ringed) remain at top of spikelet once the damage has been done and the adult has flown.

BACK TO CONTENTS

Bollworms

Helicoverpa spp.

Worldwide



Photo: K O Bell, Texas A & M University

- In addition to the damage to leaves the caterpillars attack the developing grains.
- Often a lot of frass is seen.
- These large caterpillars are never hidden within the stems or seeds.

BACK TO CONTENTS

Stemborer damage

Many species

Worldwide



Photo: A Nallusamy and C Karthikeyan, Tamil Nadu Ag University

- Exit holes in the stem often surrounded by frass.
- Almost always associated with holes in leaves.
- Cut open the stem to find the eaten cavity inside, and there may be other larvae within.
- It is necessary to see the larvae to identify the pest.
- Stems may break due to weakness caused by boreholes.

Armyworms

Spodoptera spp. and others

Worldwide



Photo: J. Mulema, CABI

- Leaves only feeders they never bore into stems.
- Larvae feed on leaf funnels, and emerging leaves will have extensive areas of damage.
- Some species feed on leaves leaving mid-rib.
- It is necessary to see the larvae to identify the pest.

BACK TO CONTENTS

Stemborer damage

Many species

Worldwide



Photo: Joseph Mulema, CABI

- Damage to the unfolded leaves result in lines of holes in the leaves once they unfold.
- Dead heart may be seen in young plants if the growing tip is eaten.

Cutworm, Turnip moth larvae

Agrotis segetum

Worldwide



Photo: W.M. Hantsbarger, Bugwood.org

- Severe damage can be done to seedlings.
- Larvae chew stems at or just above the soil line and plant the falls over.
- It looks as though the plant has been cut off.
- Larvae hide in soil during the day.

BACK TO CONTENTS

Anthracnose head rot and stalk red rot

Colletotrichum sublineolum

Worldwide



Photo: Department of Plant Pathology, North Carolina State, Bugwood.org

- Stems dark brown/black regions shiny or blotchy.
- Inside the stem is black and rotted.

Fusarium root and stalk rot

Fusarium spp.

Worldwide



Photo: L Clarfin, Kansas State University

- The lower area of the stem will appear shredded and characteristically pink or red in colour.
- Small sterile grain head is produced if anything at all.
- Isolated plants will die within the crop looking like they are suffering from drought.

BACK TO CONTENTS

Stemborer damage

Many species

Worldwide



Photo: P. Kelly, CABI

- Plants likely to fall over due to weakness in the stem.
- Plants likely to be suffering from drought stress and will mature early if they mature at all.
- Stem borer larvae are easily found once the stem is split open.

BACK TO CONTENTS

Bacterial black stalk (on maize)

Dickeya zeae

Worldwide



Photo: Howard F. Schwartz, Colorado State University, Bugwood.org

- Lower regions of stem turn brown and become water soaked soft and slimy.
- There may be a foul smell and plants frequently fall over.
- Upper leaves wilt and soft rot may occur at the base of the funnel and this may spread down rapidly. Outer leaves generally unaffected.
- Often seen in poorly drained soils.

Charcoal root and stalk rot

Macrophomina phaseolina

Worldwide



Photo: University of Nebraska, Lincoln Extension

- Stalks are soft, the plant material inside the stem is missing, only the fibers remain, small dots of fungi giving a charred “burnt” appearance.
- Plants are likely to fall over.

***Pythium* seedling and root rot**

Pythium spp.

Worldwide



Photo: A. E. Robertson, Iowa State University

- Common under prolonged waterlogged conditions.
- Seeds fail to emerge.
- Discoloured leaves develop.
- Roots rot and develop dark reddish spots, other diseases will often take over once the plant is affected.

BACK TO CONTENTS

Bacterial top and stalk rot

Dickeya zeae

Worldwide



Photo: University of Nebraska, Lincoln Extension

- The growing point of the plant disintegrates.
- Leaves are easily pulled out of the whorl and the lower end has a bad smell.
- Often basal rots accompany this symptom and the plant is likely to fall over.

Sorghum stem fly

Atherigona soccata

Africa, Asia, SE Asia



Photo: ICRISAT

- Maggot eats out the centre of the plant giving dead heart, older leaves are unaffected.
- Youngest leaf is easily removed and gives off bad smell.
- Plants may recover and send out side shoots these also may be attacked but are more able to stand the attack, papery leaves are produce in this case.

BACK TO CONTENTS

Chinch bug

Blissus leucopterus

N and S America



Photo: Gerald Wilde, CABI CPC

- Mostly attacks behind leaf sheaths but occasionally from roots and leaves.
- Severely damaged plants may die or be badly stunted (failure of leaf sheaths and stems to properly extend).
- Reddish feeding marks behind leaf sheaths and a reddish or yellowish streaking on the leaves.

Maize stripe virus

MSPV symptoms shown on Maize

Worldwide

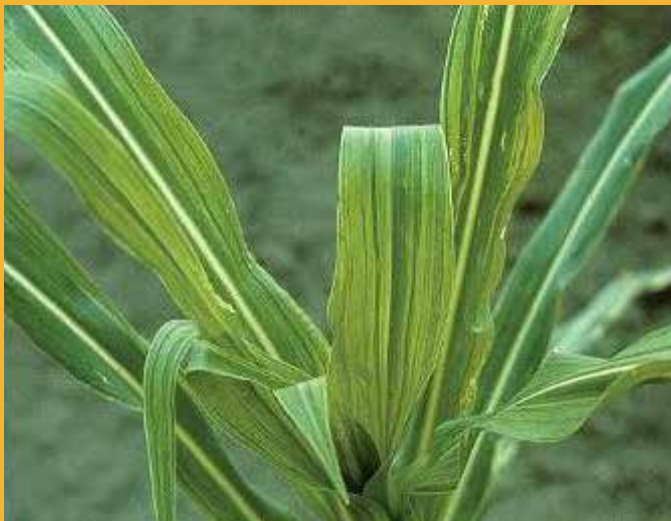


Photo: CIMMYT

- Chlorotic bands in leaves little or no panicle emerges and stunting is common.
- Chlorosis can become severe and the plant appears extremely pale.

BACK TO CONTENTS

Maize mosaic virus

MMV symptoms shown on Maize

Worldwide



Photo: CIMMYT

- Transmitted by hopper *Peregrinus maidis*.
- Discontinuous chlorotic streaks between fine veins stunting and yield loss.
- Severe if infection of young seedlings, takes place grain set reduces.

BACK TO CONTENTS

Locust and grasshopper

Locust and grasshopper

Worldwide



Photo: D Mock, Texas A&M University

- Feeding damage. Nymphs and adults feed on leaf edges, flowers and developing seed heads.
- The eaten areas are generally continuous rather than lines of holes.

BACK TO CONTENTS

Spring green aphid

Schizaphis graminum

Worldwide



Photo: P Sloderbeck and L Brooks, Texas A&M University

- Reddish yellow spots develop 2-4 days after feeding.
- These spots join together and can discolour the whole leaf which will die if feeding was severe.
- Seedlings may be killed and older plants may fall over in severe cases.

BACK TO CONTENTS

Mite damage

Mite damage

Worldwide



Photo: Kansas State University

- Extensive mite damage leads to wholesale death of leaves which gain a scorched appearance.
- Mites cannot be found on the dead leaves they are present hidden on the young leaves.
- Detailed examination of the youngest leaves is required.

BACK TO CONTENTS

Spring green aphid

Schizaphis graminum

Worldwide



Photo: B Massey, Texas A&M University

- This aphid causes much greater damage than many other kinds of aphid due to the reddening leaf spots that develop where feeding takes place.
- These areas can become so extensive that whole plants can die.
- Close inspection of the plants are required.

BACK TO CONTENTS

Anthracnose

Colletotrichum sublineolum

Worldwide

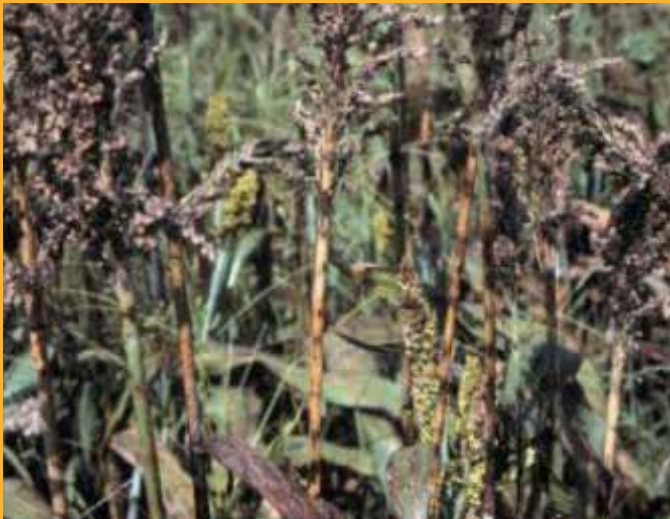


Photo: Uni Georgia Dept Agricultural and Environmental sciences

- Spots on the stem are circular with a red to black border and grey centre. They penetrate into the stem and red marbling can be seen inside the stem.
- Fruiting bodies often seen within the spots on the stems.
- The disease can kill crops that have poor resistance.

BACK TO CONTENTS

Rust

Puccinia purpurea

Worldwide



Photo: S Nelson, Flickr

- The leaves bearing the reddish powdery pustules will dry up.
- Plants will show poor head development and this will lead to shrivelled grain.
- Under ideal conditions the whole plant will be affected.

Sting nematode

Belonolaimus spp.

N America



Photo: Clemson University USDA, Bugwood.org

- Seedlings especially affected.
- Very reduced root system with short stubby roots with black areas on them especially at the tips.
- As roots are eaten new ones develop creating a highly branched root system.
- Plants may grow away from damage with time.

Stubby root nematode

Paratrichodorus minor

Worldwide

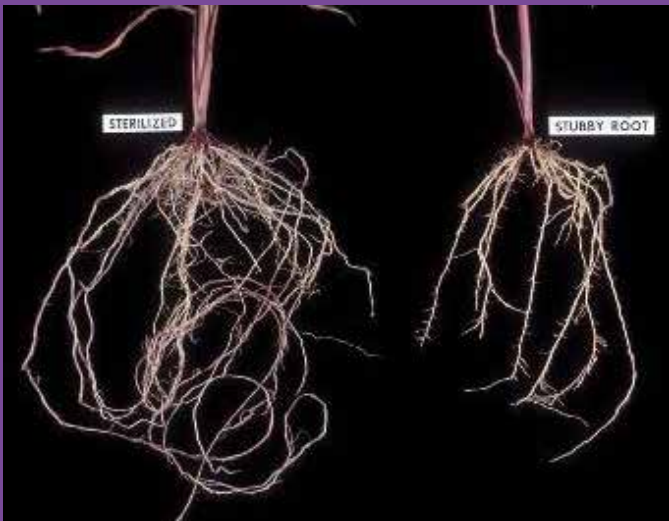


Photo: J Dept, Plant path NCSU, Bugwood

- Seedlings especially affected.
- Very reduced root system with short stubby roots with black areas on them especially at the tips.
- As roots are eaten new ones develop creating a highly branched root system.
- Plants may grow away from damage with time.

BACK TO CONTENTS

Lesion nematode

Pratylenchus spp. (on banana)

Worldwide



Photo: University of Nebraska

- Causes black dead areas on root surface and holes inside roots.
- Above ground only general symptoms develop, reduced growth, with stunting.

BACK TO CONTENTS

Root knot nematode

Meloidogyne spp.

Worldwide



Photo: Dept of Plant Path, NCSU, Bugwood.org

- Plants, small, yellow less vigorous. Looks like drought and nutrient deficiencies.
- Symptoms are particularly severe when plants are infected soon after planting.
- Below ground galls ranging in size from 1-10 mm. In severe infestations, heavily galled roots may rot away, leaving a poor root system with a few large galls.

BACK TO CONTENTS

Stunt nematode

Tylenchorhynchus spp.

Worldwide



Photo: Dept of Plant Path, NCSU, Bugwood.org

- Root feeder. Few stubby roots, swollen brown tips.
- Plants, small, yellow, less vigorous.
- Looks like drought and nutrient deficiencies.
- Makes other root diseases such as charcoal rot more likely to occur.

BACK TO CONTENTS

Witchweed

Striga asiatica

Africa and Asia



Photo: P. Taylor, CABI

- Parasite or hemi-parasitic reddish flowered plant causes scorching early drying and often the plant will fall.
- Plants will show poor vigour and fail to yield.

BACK TO CONTENTS

Witchweed

Striga hermonithica

Africa



Photo: IITA image library

- Parasite or hemi-parasitic purple flowered plant causes scorching early drying and often the plant will fall.
- Plants will show poor vigour and fail to yield.

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